Web Mining and Recommender Systems

Ethics of Recommendation and Personalization

Learning Goals

- Discuss how recommender systems (and other personalized algorithms) can lead to unfair outcomes
- Discuss general strategies to mitigate these outcomes
- Explore various case-studies about the dangers of recommendation

What are fairness issues in *personalized* algorithms?

- Recommender systems may have a 'concentration' effect, where users are gradually locked into a 'filter bubble' containing only a narrow set of items
- Recommending content maximally aligned with a user's interests may gradually push users toward more and more 'extreme' content
- Recommender systems may have reduced utility for users (or groups of users) who are underrepresented in the training data
- Recommendations may focus only a user's predominant interest, while failing to capture the diversity and breadth of their interactions
- Systems could disadvantage *vendors* (or content creators, etc.) by failing to recommend products in the long-tail

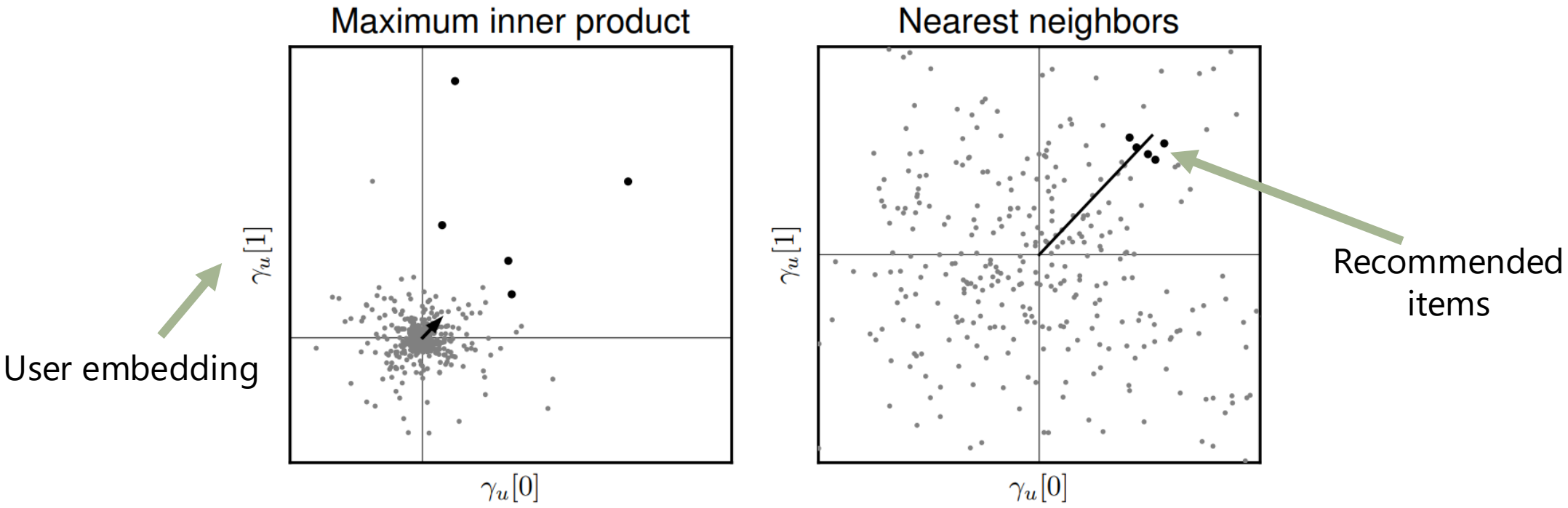
What problematic outcomes could recommenders have?

Consider recommending items by taking the maximum inner product versus a nearest neighbor

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What problematic outcomes could recommenders have?



Consider recommending items by taking the maximum inner product versus a nearest neighbor

$$rec(u) = \arg\max_{i \in I \setminus I_u} f(u, i)$$

$$f(u, i) = \|\gamma_u - \gamma_i\|$$

What problematic outcomes could recommenders have?

- Maximizing the inner product will tend to recommend "extreme" items: if I like action, I should like *a lot* of action
- Finding nearest neighbors will tend to recommend items that are very close to what I've already consumed

- **How can we measure (and maybe correct) these issues with content extremity / diversity?**

Diversity in recommendation

Let's try to compare users' *consumption* histories to what gets *recommended*

E.g. compute how many times each item is recommended:

```
countsPerItem = defaultdict(int)

for u in range(nUsers):
  # Given a matrix of interactions
  recs = model.recommend (u, Xui, N=len(itemsPerUser [u]))
  for i, score in recs:
    countsPerItem [i] += 1
```

Diversity in recommendation

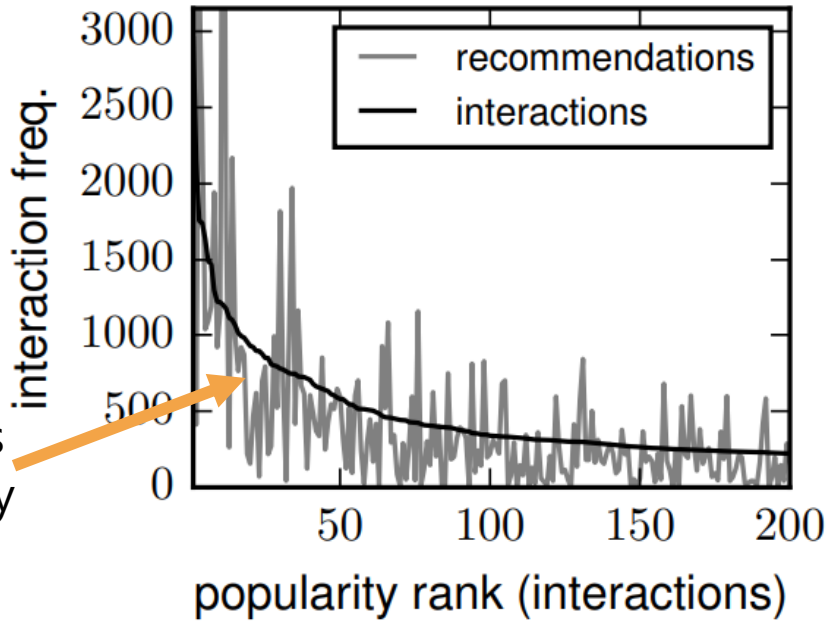
Questions:

1. **Are items that were consumed a lot the same as the ones that tend to be recommended a lot? How well do consumptions and recommendation distributions overlap?**
2. What about the shape of the distribution? Are recommendations dominated by popular items (moreso than consumptions?)

Diversity in recommendation

Consumption vs recommendation distribution (Goodreads)

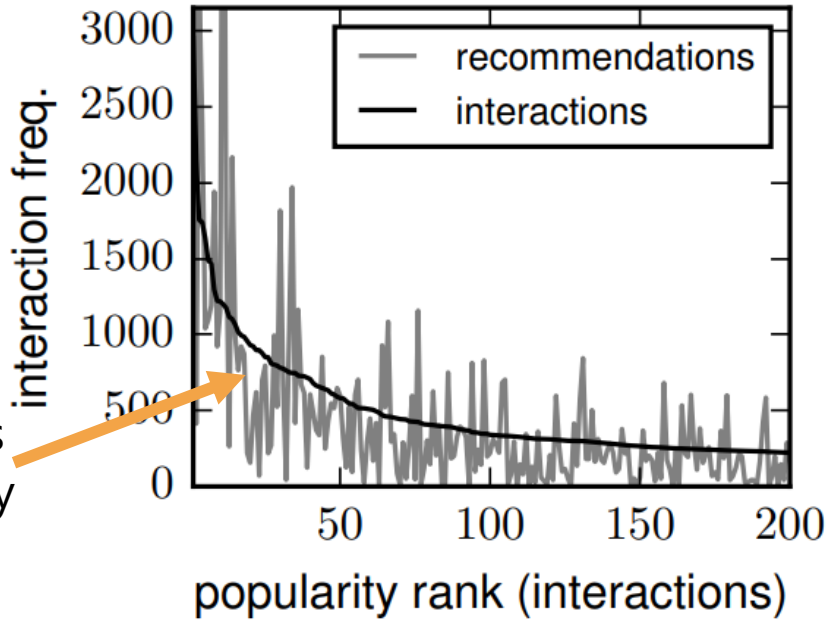
The most interacted items tend to be highly recommended

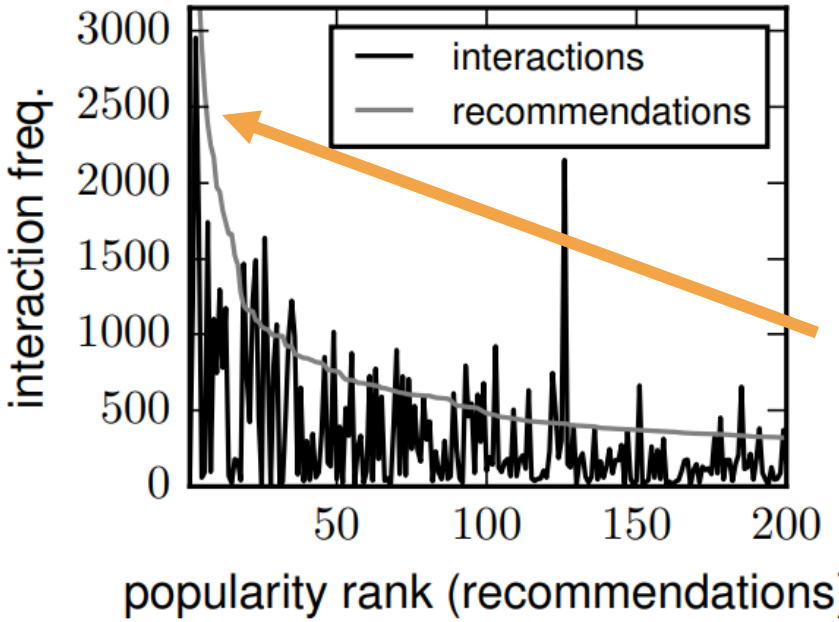


Diversity in recommendation

Consumption vs recommendation distribution (Goodreads)

The most interacted items tend to be highly recommended





The most recommended items had many interactions

Distributions match okay, but top-recommended items appear *much* more than their number of historical interactions!

Diversity in recommendation

Questions:

1. Are items that were consumed a lot the same as the ones that tend to be recommended a lot? How well do consumptions and recommendation distributions overlap?
2. **What about the shape of the distribution? Are recommendations dominated by popular items (moreso than consumptions?)**

Concentration

We saw in the previous example that the most-recommended items were a fair bit more popular than the most consumed items

*i.e., the recommender made popular items **more popular***

This is known as a **concentration effect:** recommendations may concentrate around a few items (and this could cause a feedback loop!)

Measuring concentration

We can measure concentration via the *Gini Coefficient*:

$$G(y) = \frac{\sum_{i=1}^{N} \sum_{j=1}^{N} |y_i - y_j|}{2N^2 \bar{y}}.$$

The Gini coefficient measures the average difference items in a set, e.g. the average difference in wealth between individuals

- Close to zero: everyone has about the same wealth (uniform)
- Close to 1: wealth is concentrated among a few individuals

Measuring concentration

For a recommender system, we might be interested in the *difference* between Gini coefficients of interactions versus recommendations

- If recommendations have a *higher* Gini coefficient than interactions, then the recommender is causing a concentration effect
- If recommendations have a *lower* Gini coefficient than interactions, then the recommender is causing a dispersion (or diversification?) effect

Measuring concentration

Measuring the Gini coefficient

```
def gini(y, samples =1000000)
  m = sum(y) / len(y) # average
  denom = 2 * samples * m
  numer = 0
  for _ in range(samples):
    i = random.choice(y)
    j = random.choice(y)
    numer += math.fabs(i - j)

return numer / denom
```



Measured on a sample for a large corpus

For Goodreads: interactions have G = 0.72; recommendations have G = 0.77, i.e., slight concentration

Measuring concentration & diversity

From some real studies:

- **Fleder and Hosanagar, 2009:** Simulate users (can accept or reject recommendations), with recommenders trained on interaction history. Over time, recommendations become more and more concentrated
- **Nguyen et al. 2014:** For real users, both recommendations and interactions become less diverse over time (in terms of content features)

Measuring concentration & diversity

(See more in textbook)

- **Extremification (Ribeiro 2020, youtube):** How do recommendations on youtube guide users to extreme content? E.g. if users visit pages that have a specific slant (but are not "extreme"), will they gradually be guided to more extreme pages?
- **Content diversity (Zhou 2010, youtube):** Recommendations drive a large fraction of views, and are more diverse than what would be expected by popularity-driven models

References

Further reading:

- Nguyen et al. 2014. Exploring the filter bubble: The effect of using recommender systems on content diversity. In: International World Wide Web Conference
- Fleder, Daniel, and Hosanagar, Kartik. 2009. Blockbuster culture's next rise or fall: The impact of recommender systems on sales diversity. Management Science

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Fairness issues in recommendation: algorithmically correcting concentration/diversity issues

Re-ranking strategies to diversification

A simple way to make recommendations more "diverse" is just to (post-hoc) *rerank* the outputs of some recommender

Note that diversity could mean a few things:

1. Is there variety among the **set of items** a user is recommended?
2. Across **all users,** are different items recommended to different people?

Maximal marginal relevance (MMR)

1. Is there variety among the **set of items** a user is recommended?

Basic strategy (Carbonell and Goldstein, 1998):

- Start with the most relevant item
- Repeatedly select the next most relevant item, but penalize relevance if it's too similar to already selected items
- Repeat until we have the desired number of items

Maximal marginal relevance (MMR)

1. Is there variety among the **set of items** a user is recommended?

Basic strategy (Carbonell and Goldstein, 1998):

Maximal marginal relevance (MMR)

1. Is there variety among the **set of items** a user is recommended?

Basic strategy (Carbonell and Goldstein, 1998):

$$MMR = \arg\max_{i \in R \setminus S} \left[\lambda \underbrace{Sim^{\text{user}}(i, u)}_{\text{relevance to the user}} - (1 - \lambda) \overbrace{\max_{j \in S} Sim^{\text{item}}(i, j)}^{\text{similarity to already-recommended items}} \right],$$

Large lambda: only care about relevance

Small lambda: only care about diversity

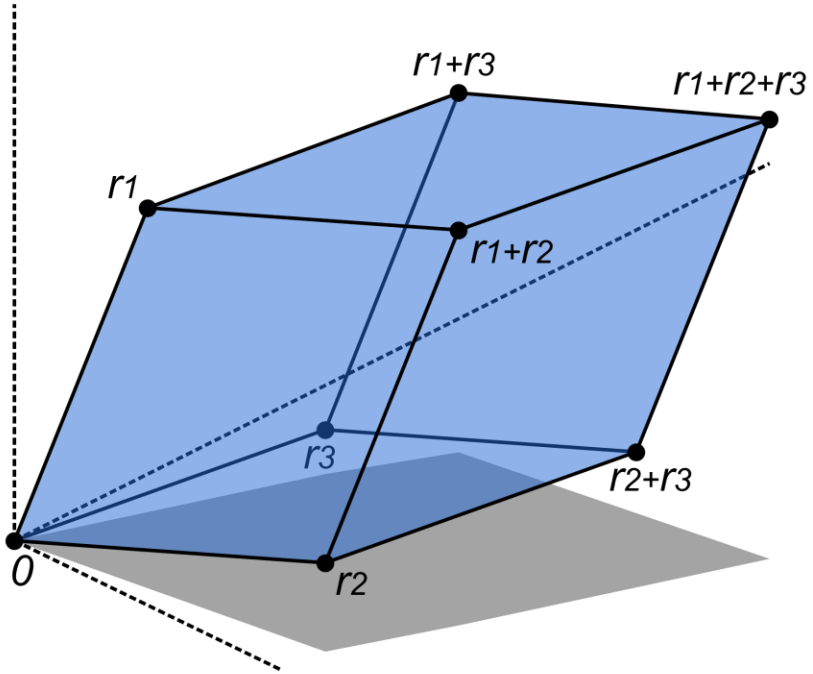
Maximal marginal relevance (MMR)

Examples (beer recommendations):

Low diversity	Medium diversity	High diversity
Founders KBS (Kentucky Breakfast Stout)	Founders KBS (Kentucky Breakfast Stout)	Founders KBS (Kentucky Breakfast Stout)
Two Hearted Ale	Samuel Smith's Nut Brown Ale	Samuel Smith's Nut Brown Ale
Bell's Hopslam Ale	Two Hearted Ale	Salvator Doppel Bock
Pliny The Elder	Bell’s Hopslam Ale	Oil Of Aphrodite - Rum Barrel Aged
Samuel Smith's Oatmeal Stout	Kolsch	Great Lakes Grassroots Ale
Blind Pig IPA	Drax Beer	Blue Dot Double India Pale Ale
Stone Ruination IPA	A Little Sumpin' Extra! Ale	Calistoga Wheat
Schneider Aventinus	Odell Cutthroat Porter	Dogwood Decadent Ale
The Abyss	Miner's Daughter Oatmeal Stout	Traquair Jacobite
Northern Hemisphere Harvest Wet Hop Ale	Rare Bourbon County Stout	Cantillon Gueuze 100% Lambic

Determinantal Point Processes (DPPs)

1. Is there variety among the **set of items** a user is recommended?



$$\det\left(\begin{bmatrix} L_{i,i} & L_{i,j} \\ L_{j,i} & L_{j,j} \end{bmatrix}\right) = L_{i,i}L_{j,j} - L_{i,j}L_{j,i}$$

Off diagonal: similarity between two items

Diagonal: relevance of an item

Determinantal Point Processes (DPPs)

1. Is there variety among the **set of items** a user is recommended?

Basic strategy (Kulesza and Taskar, 2012)

- Want to select a set of items with high determinant
- In practice this is hard, so approach is the same as with MMR (i.e., just iteratively select items to incrementally increase the determinant)

Re-ranking strategies to diversification

2. Across **all users,** are different items recommended to different people?

Alternate view: can we recommend things to people that are relevant but not "obvious" (see: serendipity, later)

Re-ranking strategies to diversification

2. Across **all users,** are different items recommended to different people?

Strategy (from n Adomavicius and Kwon, 2011):

- Replace an item's original rank (relevance) with

Popularity (e.g. number of historical interactions)

Only if relevance is high enouh

$$rank'_u(i,t) = \begin{cases} rank^{(\text{pop})}(i) & \text{if } r(u,i) \geq t \\ \alpha_u + rank_u(i) & \text{otherwise} \end{cases}$$

Original rank (relevance)

Re-ranking strategies to diversification

2. Across **all users,** are different items recommended to different people?

In other words, *recommend items that I like, but which aren't popular in general* (remember: tf-idf)

This will spread recommendations across less-popular items and (maybe?) help with discovery

Re-ranking strategies to diversification

Note: nothing here specific to recommendation – these types of diversification strategies could work for any ranking algorithm (e.g. MMR predates this type of recommender system altogether)

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Case studies: Diversity and Filter Bubbles

Spotify (Anderson et al. 2020)

Idea: Some users ("generalists") may prefer diverse recommendations, while others ("specialists") may prefer narrower recommendations

Q: How do different types of user respond to diverse recommendations (on Spotify)?

Spotify (Anderson et al. 2020)

Model is an item2vec style model (see text mining lectures) that produces item representations \gamma_I

The user representation is just the average of their listening history:

$$\gamma_u = \frac{1}{|H|} \sum_{j=1}^{|H|} \gamma_{H_j},$$

Spotify (Anderson et al. 2020)

A **specialist** is a user whose consumption patterns tend to be oriented in a particular direction

A **generalist** is a user whose consumptions are not oriented in any particular direction

This is measured by cosine similarity:

$$GS(u) = \frac{1}{|H|} \sum_{j=1}^{|H|} \frac{\gamma_{H_j} \cdot \gamma_u}{\|\gamma_{H_j}\| \|\gamma_u\|}$$

Spotify (Anderson et al. 2020)

$$GS(u) = \frac{1}{|H|} \sum_{j=1}^{|H|} \frac{\gamma_{H_j} \cdot \gamma_u}{\|\gamma_{H_j}\| \|\gamma_u\|}$$

Specialist (GS close to 1): all listens have similar representations
Generalist (GS close to 0): listens more "spread out"

How does this score relate to patterns of user activity?

- Inactive users tend to be specialists (GS close to 1)
- Generalists (GS close to 0) more likely to "churn"

Spotify (Anderson et al. 2020)

How do different respond differently to recommendations?

- Both groups are more engaged by recommendations (versus popularity alone)
- But specialists benefit more from recommendations (more significant boost in engagement), and are more sensitive to songs matching their personal relevance criteria

Filter Bubbles

How does personalization affect content diversity?
(really, are users trapped in filter bubbles?)

1. **Do recommendations guide users to less diverse content over time?**
2. Do recommender systems guide users to extreme content?
3. Are online recommenders worse than "traditional" forms of online media?

Content diversity over time (Bakshy et al. 2015)

Bakshy et al. (2015) studied the extent to which recommendations on *facebook* reinforce users' political ideology

Main questions:

1. How much do recommenders **expose** users to diverse content?
2. How much do users **interact with** cross-cutting content?

Content diversity over time (Bakshy et al. 2015)

1. How much do recommenders **expose** users to diverse content?

Various challenges: e.g. labeling content ("liberal"/"conservative"/"neutral"), and dealing with exposure bias (e.g. liberal people have liberal friends, so are exposed to liberal content), so naturally most content is not cross-cutting

Content diversity over time (Bakshy et al. 2015)

1. How much do recommenders **expose** users to diverse content?

- Algorithmic ranking indeed exposes users to less ideologically diverse news than would be expected by the ideological makeup of their social group

Content diversity over time (Bakshy et al. 2015)

2. How much do users **interact with** cross-cutting content?

- Users interact with ideologically diverse content at an even lower rate than their rate of exposure.
- Arguably, individual choice plays the largest role in users' exposure to content that is ideologically homogeneous (according to facebook!)

Filter Bubbles

(More on other two filter bubble / extremification issues in textbook)

1. Do recommendations guide users to less diverse content over time?
2. **Do recommender systems guide users to extreme content?**
3. **Are online recommenders worse than "traditional" forms of online media?**

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Other Metrics Beyond Accuracy

Desirable features of a recommender

We've focused on *diversity* as our main metric (other than accuracy/relevance). What other features are desirable?

- Items should be **novel,** i.e., we should balance discovery of new items against recommending items with high interaction probability (but which are already known)
- Rather than being internally diverse, we might have goals such as **mutual compatibility** among items (see e.g. outfit generation)
- Recommended items should have good **coverage,** i.e., they should represent a broad range of categories or features; or they should be **balanced,** in terms of matching the category distribution from the user's history
- Other goals could be more nebulous, such as perceived **unexpectedness, serendipity,** or overall user **satisfaction**

1. Serendipity

Idea: Recommendations should be "novel" to the user in the sense that they aid discovery, i.e., they should be *serendipitous*

What is serendipity?

1. Serendipity

- *Unexpectedness* to users (didn't expect to be relevant or recommended) (Kotkov et al. 2018)
  - Low *popularity*
  - Differs across demographics (e.g. male/female old/young users may perceive recommendations as more or less serendipitous) (Wang et al. 2020)
  - May have to do with familiarity with the platform

(not clear what "serendipity" is, but is a combination combination of relevance, novelty, and unexpectedness)

Serendipity

How similar is a recommended song compared to songs in a user's interaction history (Zhang et al. 2012)?

$$\text{Unserendipity} = \frac{1}{|U|} \sum_{u \in U} \frac{1}{|I_u|} \sum_{i \in I_u} \sum_{j \in R_u} \frac{\text{Cos}(i, j)}{|R_u|}$$

Average cosine similarity

Historical interactions by *u*

Recommendations for *u*

(*un*serendipity, so high values = not serendipitous)

Serendipity

What is the optimal tradeoff between

- *Serendipity*
- *Diversity* (e.g. cosine similarity among recommendations)
- *Novelty* (e.g. item popuarity)

Serendipity

What is the optimal tradeoff?

- Items a user didn't expect to be relevant can negatively impact satisfaction (Kotkov et al. 2018)
- Serendipity broadens preferences, but doesn't impact satisfaction (Kotkov et al. 2018)
- Results can be perceived as serendiptous even when similar recommendations have been received recently (Wang et al. 2020)

2. Calibration (Steck, 2018)

Idea: Recommendations should have similar attribute proportions to my past interactions

E.g. if I watched 80% romance and 20% comedy on Netflix, my recommendations should not be 100% romance

2. Calibration (Steck, 2018)

First, define a probabilistic attribute vector for each item $p(g|i)$

E.g. Harry Potter might be 10% romance, 5% comedy, 20% action, 50% fantasy (etc.)

Distribution of all recommended genres should match historical genre consumption

2. Calibration (Steck, 2018)

Second, measure recommended versus historical genre distribution for all items consumed/recommended for a user *u*:

$$\text{historical:} \quad p(g|u) = \frac{\sum_{i \in I_u} w_{u,i} \cdot p(g|i)}{\sum_{i \in I_u} w_{u,i}}$$

$$\text{recommended:} \quad q(g|u) = \frac{\sum_{i \in R_u} w_{r(i)} \cdot p(g|i)}{\sum_{i \in R_u} w_{r(i)}}$$

2. Calibration (Steck, 2018)

Second, measure recommended versus historical genre distribution for all items consumed/recommended for a user *u*:

consumed items

$$\text{historical:} \quad p(g|u) = \frac{\sum_{i \in I_u} w_{u,i} \cdot p(g|i)}{\sum_{i \in I_u} w_{u,i}}$$

(optional) weighting, e.g. by recency

$$\text{recommended:} \quad q(g|u) = \frac{\sum_{i \in R_u} w_{r(i)} \cdot p(g|i)}{\sum_{i \in R_u} w_{r(i)}}$$

recommended items

2. Calibration (Steck, 2018)

The goal is that the two distributions should (approximately) match

(they use the KL divergence)

$$\mathrm{KL}(p, q) = \sum_{g} p(g|u) \log \frac{p(g|u)}{q(g|u)}$$

2. Calibration (Steck, 2018)

In practice recommendation is the same as with our diversity approaches, i.e., iteratively add new recommendations that balance compatibility and calibration

$$R_u = \arg\max_R \underbrace{(1-\lambda) \cdot \sum_{i \in R} f(u,i)}_{\text{compatibility}} - \underbrace{\lambda \cdot \mathrm{KL}(p, q(R))}_{\text{calibration}}$$

Small lambda: only care about relevance

Large lambda: should match historical distribution very closely

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Fairness

Background: fairness in Machine Learning (not recommendation)

The dangers of "blindly" applying machine learning to biased datasets are receiving increasing attention. For example:

- If hiring practices were historically sexist, then an ML algorithm trained on hiring outcomes may propagate sexist decisions
- If a facial recognition system was trained on a predominantly white population, it may have poor performance on other races
- Not just data: a choice of regularization strategy or error metric could favor "normal" samples versus outliers

Background: fairness in Machine Learning

Given these issues of bias, a classic setting of "fair ML" consists of building debiased classifiers

Suppose we are given a dataset with some label ***y*** (such as being qualified for a job), and we are concerned that decisions may be biased against individuals exhibiting a sensitive attribute (from a feature vector ***x***)

Background: fairness in Machine Learning

Example: demographic parity

$$P(f(x_i) = 1 | x_{i,f} = 1) = P(f(x_i) = 1 | x_{i,f} = 0)$$

Predicted label

Value of sensitive attribute

- Percentage of candidates *predicted as qualified* should be equal among both groups
- Can be achieved (for instance) by **balancing** our dataset (Kamiran & Calders, 2009)

Background: fairness in Machine Learning

Example: equal opportunity

$$P(f(x_i) = 1 | x_{i,f} = 1, y_i = y) = P(f(x_i) = 1 | x_{i,f} = 0, y_i = y)$$

Predicted label

Value of sensitive attribute

Label

- Among qualified (or unqualified) individuals, the probability of prediction should be the same regardless of the protected feature

Questions from fair ML

- Can we resample our datasets so that they correct for imbalance, and will lead to fair classifiers? (Kamiran & Calders, 2009)
- Can we change the classifier itself so that it will make fair decisions? Can we do so without using the protected attribute? (Zafar et al., 2017)
- What are the unintended consequences of "fair" classification techniques? (Lipton et al., 2018)

Questions from fair ML

- Fairness measures need not be thought of as constraints that should be *enforced;* rather, we might want to **make sure that algorithms don't exacerbate existing patterns of bias**

References

Further reading:

- Classifying without discriminating. Kamiran, Calders, 2009
- Fairness Beyond Disparate Treatment & Disparate Impact. Zafar, Valera, Gomez Rodriguez, Gummadi, 2017
- Does mitigating ML's impact disparity require treatment disparity? Lipton, Chouldechova, McAuley, 2018

Web Mining and Recommender Systems

Case studies: Gender bias in recommendation

How is fairness in recommendation different?

- Recall we're optimizing something like:

$$\arg\min_{\alpha,\beta,\gamma} \sum_{u,i} (\alpha + \beta_u + \beta_i + \gamma_u \cdot \gamma_i - R_{u,i})^2$$

- I.e., we're making personalized recommendations to each user based on some model

- In what ways could such a model be problematic, unbiased, or unfair?

How is fairness in recommendation different?

- Yao & Huang (2017) studied fairness in a recommendation setting based on online course evaluations of CS classes
  - Model is of the form

$$\arg\min_{\alpha,\beta,\gamma} \sum_{u,i} (\alpha + \beta_u + \beta_i + \gamma_u \cdot \gamma_i - R_{u,i})^2$$

- **What might happen if females (or any group) is highly underrepresented in this type of data?**

How is fairness in recommendation different?

E.g. the underrepresented group might have their ratings over or underpredicted ("value unfairness")

$$U_{\text{val}} = \frac{1}{|I|} \sum_{i=1}^{|I|} \left| \left(\underbrace{\mathbb{E}_g[y]_i}_{\text{expected } \textit{prediction} \text{ for group } g \text{ on item } i} - \overbrace{\mathbb{E}_g[r]_i}^{\text{average } \textit{rating} \text{ for group } g \text{ on item } i} \right) - \left(\mathbb{E}_{\neg g}[y]_i - \mathbb{E}_{\neg g}[r]_i \right) \right|$$

Misprediction for males

Misprediction for females

(equation is a mouthful but it's just the *difference of mispredictions* for the two groups)

How is fairness in recommendation different?

We could measure related quantities in various ways:

$$U_{\text{abs}} = \frac{1}{|I|}\sum_{i=1}^{|I|}\left|\left|\mathbb{E}_g[y]_i - \mathbb{E}_g[r]_i\right| - \left|\mathbb{E}_{\neg g}[y]_i - \mathbb{E}_{\neg g}[r]_i\right|\right|$$

$$U_{\text{under}} = \frac{1}{|I|}\sum_{i=1}^{|I|}\left|\max\{0, \mathbb{E}_g[r]_i - \mathbb{E}_g[y]_i\} - \max\{\mathbb{E}_{\neg g}[r]_i - \mathbb{E}_{\neg g}[y]_i\}\right|$$

$$U_{\text{over}} = \frac{1}{|I|}\sum_{i=1}^{|I|}\left|\max\{0, \mathbb{E}_g[y]_i - \mathbb{E}_g[r]_i\} - \max\{\mathbb{E}_{\neg g}[y]_i - \mathbb{E}_{\neg g}[r]_i\}\right|$$

Disparity in recommendation

Ultimately each is a form of **disparity** between the two groups

First main point is that such disparities are manifest in real datasets with standard recommendation approaches (most experiments are on movie recommendation, across categories that exhibit different levels of gender imbalance)

Disparity in recommendation

Second point is that these fairness objectives can be incorporated into training with little loss in performance:

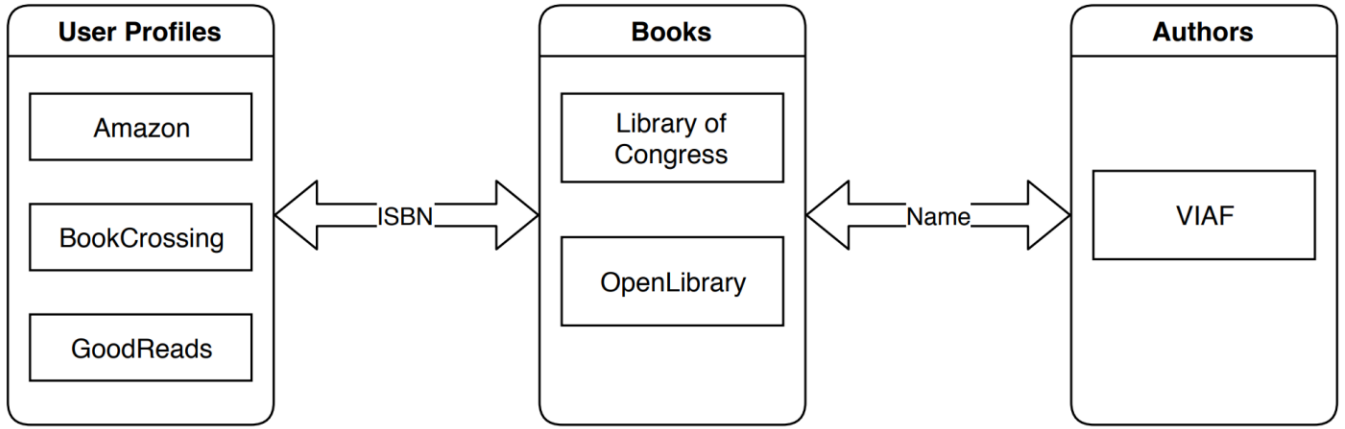
$$\frac{1}{\mathcal{T}} \sum_{(u,i)\in\mathcal{T}} \underbrace{(\alpha + \beta_i + \beta_u + \gamma_i + \gamma_u - R_{u,i})^2}_{\text{accuracy}} + \lambda \underbrace{U_{\text{abs}}}_{\text{(absolute) fairness}}.$$

Web Mining and Recommender Systems

Case study 2: Bias in author gender

Gender bias in book recommendations

- Ekstrand and Kluver (2019) explored gender in book recommendations, on *Amazon*, *BookCrossing*, and *GoodReads*
- Review data is available, but some considerable effort to extract author gender



Gender bias in book recommendations

- Various research questions:

1/2: To what extent are female authors over/underrepresented in the dataset (1) and among users' consumption patterns (2)

3/4: To what extent do recommenders mimic or exacerbate any imbalance?

5: Can this be algorithmically corrected, and what is the cost in doing so?

Aside: C-, P-, and CP-fairnes

In recommender systems, fairness can be viewed from the perspective of the **consumer (C),** the **producer (P),** or **both (CP)** (Burke, 2017)

- Previous paper: C-fairness
- This paper: P-fairness
- Next paper: CP-fairness

RQ 1/2 (gender in data and interactions):

Male-authored books are overrepresented; less so in interactions (and in fact less than an ostensible distribution of all authors); individual users are quite diverse

	Books		Ratings	
Data	female	male	female	male
LOC	22.7%	77.3%	—	—
AZ	30.6%	69.4%	38.9%	61.1%
BX-E	40.5%	59.5%	43.0%	57.0%
BX-I	40.7%	59.3%	45.7%	54.3%
GR-E	37.8%	62.2%	47.6%	52.4%
GR-I	37.7%	62.3%	48.2%	51.8%

Library of congress → LOC

Amazon → AZ

BookCrossing ratings → BX-E

all interactions → BX-I

GoodReads → GR-E

RQ 3/4 (distribution after recommendation):

Several standard recommendation approaches are considered

- *Implicit* models make use of interactions
- *Explicit* models make use of ratings

- Mostly, implicit models preserve users' historical gender skew
- Some explicit models propagate the overall skew of the data (i.e., toward male authors)

RQ 5 (algorithmic correction):

Rather than correcting this bias during training (like in Yao & Huang), bias is corrected using a post-hoc reranking strategy

Strategy(ies) are simple greedy algorithms that recommend items with high utility while enforcing a balance constraint (note: this is what might be called "affirmative action" in the fairness literature)

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Case study 3: Bias in marketing images

Gender/congruency bias in marketing

Motivating question:

When I buy products, how much am I influenced by whether models who market the product look "like me" (gender, race, body type, etc.)? (Also called "self congruency")



Addressing Marketing Bias in Product Recommendations, *WSDM 2020*

Gender/congruency bias in marketing

Potential fairness issues:

- Users poorly represented by marketing may struggle to find products they like
- Vendors may miss out on sales by mis-marketing their products

This is an example of **multisided** fairness in recommendation (see e.g. Burke, 2017)

Gender bias in marketing

Main research questions are similar to previous ones:

RQ1: Do users follow self-congruency when selecting items? (even for items where we might expect this to be irrelevant)

RQ2: Does this lead to fairness issues in recommended items?

RQ3: Can this be algorithmically mitigated?

Gender/congruency bias in marketing

Main problem is actually dataset construction…

(1) ModCloth clothing data. ModCloth explicitly states body type of models (small, plus, etc.)



Can separate users into groups based on which body types they buy; and we can find items that are *available* in multiple types but modeled using a specific type

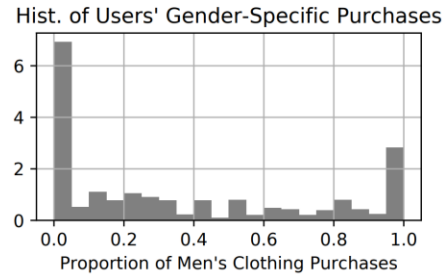
Gender/congruency bias in marketing

Main problem is actually dataset construction…

(2) Amazon Electronics data



Use an off-the-shelf detector to determine gender in marketing images (Face++); determine user "gender" based on their purchases in clothing categories



Gender/congruency bias in marketing

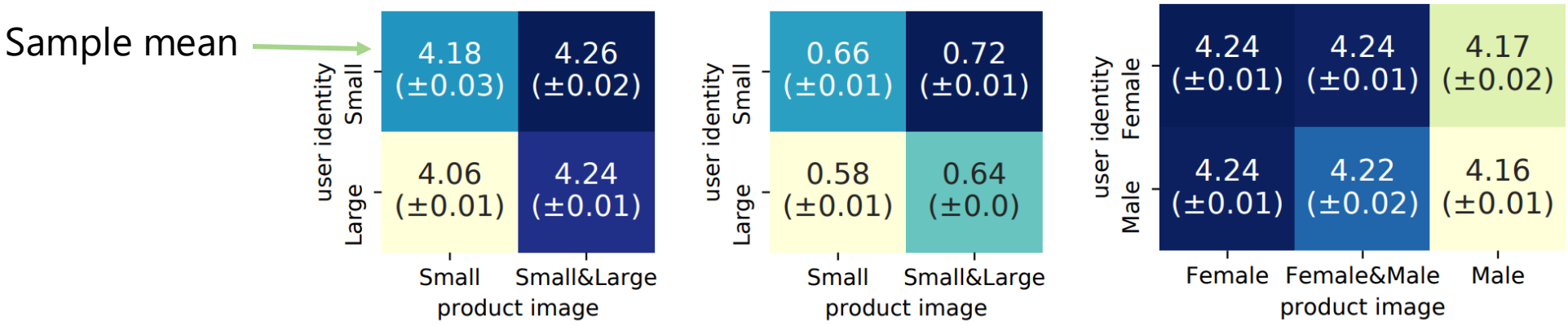
Note: lots of **big** assumptions being made here!

(E.g.) clothing marketed with a plus-size model may not map well to a small user, even if a small version is available

The points are to (a) **measure** whether self-congruency bias exists; (b) to determine whether it's **propagated** by recommendations; and (c) to algorithmically correct it **in any specific cases where we might want to**

RQ1: Does self-congruency hypothesis apply?

Users interact with / give higher ratings to products that are marketed specifically to their group



(again, this doesn't necessarily point to any fairness issue)

RQ2: Do recommender systems propogate bias?

Users who buy products *not* marketed to them receive lower utility (higher error) from recommendations (F-test)

$$\text{Product Image} \begin{cases} \text{Female} \\ \text{Male} \end{cases} \underbrace{\begin{bmatrix} \bar{e}_{F,F} & \bar{e}_{M,F} \\ \bar{e}_{F,M} & \bar{e}_{M,M} \end{bmatrix}}_{\substack{\text{Female} \quad \text{Male} \\ \text{User Identity}}}$$

(this potentially *is* a fairness issue)

RQ3: Can this be algorithmically corrected?

This is corrected using a similar strategy to what we saw previously:

$$\underbrace{\sum_{u,i}(f(u,i)-r_{u,i})^2}_{\text{prediction error}} + \alpha \overbrace{\mathcal{L}_{corr}}^{\text{error parity on market segments}}.$$

Again, results show error parity can be achieved with little loss in utility (and sometimes a gain in utility!)

Summary

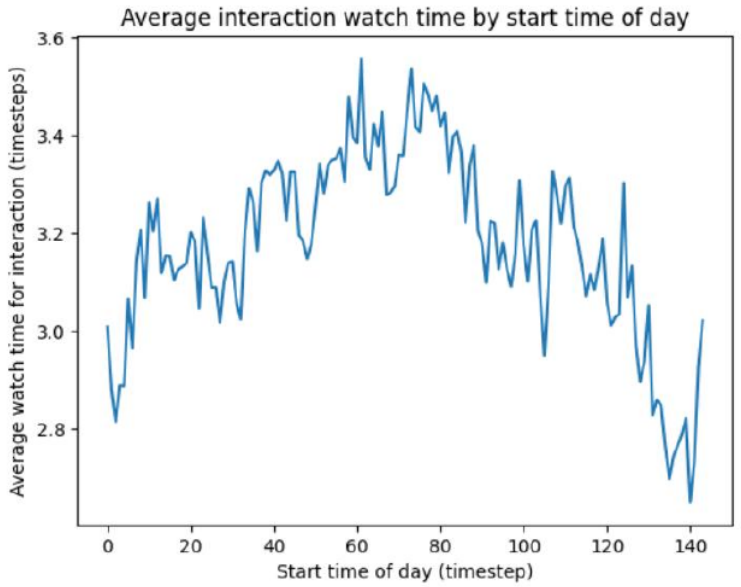
- Fairness in recommendation has quite different metrics / goals than traditional fairness problems
- Most have to do with loss of utility for certain groups, and ensuring that recommenders don't make things **worse**
- Straightforward correction strategies that balance fairness objectives with recommendation utility
- Topic is still quite new and open!

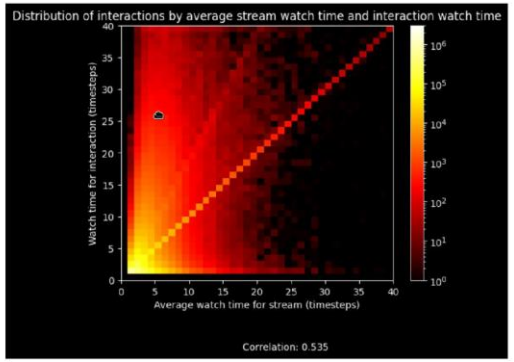
- Lots of other related perspectives: e.g. calibration, filter bubbles, content diversity, extremification

Web Mining and Recommender Systems

Assignment 2 Submissions

Watch time prediction for Twitch streams





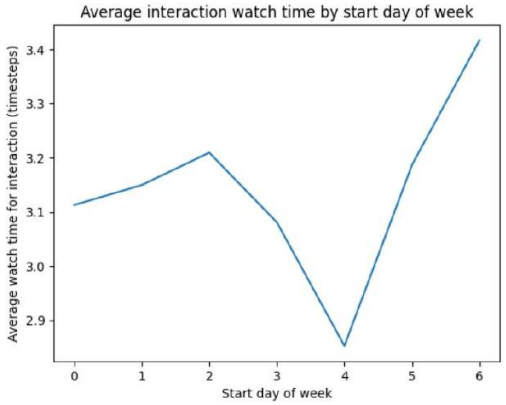


Figure 5: Distribution of interactions by average stream watch time and interaction watch time. Each (x,y) displays how many interactions lasted for y timesteps and was part of a stream whose watch times averaged x timesteps. Truncated to 40 timesteps due to low frequency of other data.

Geographical recommendation

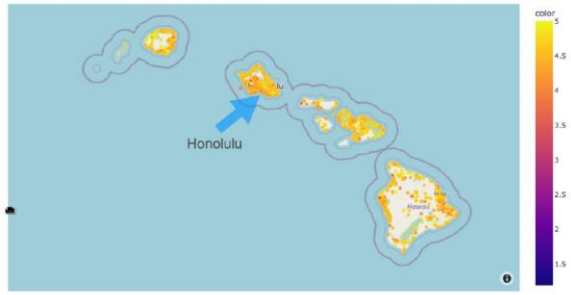


Figure 3: Ratings among the whole Hawaii area

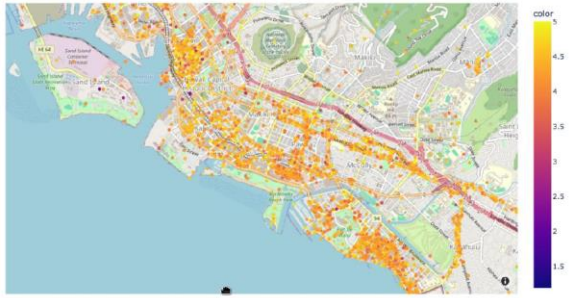


Figure 4: Ratings among the Honolulu area

Surviving the Titanic

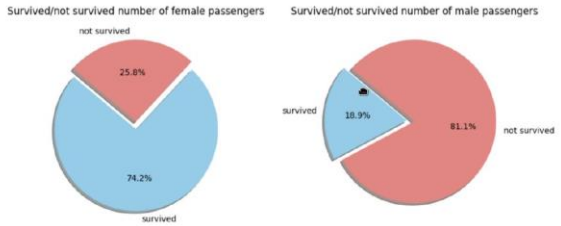
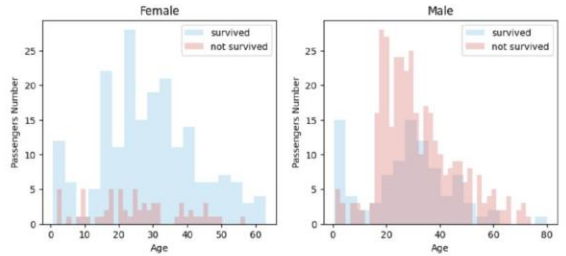
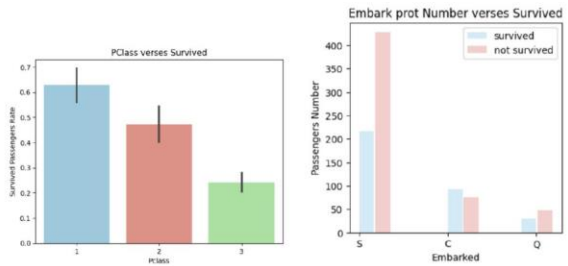


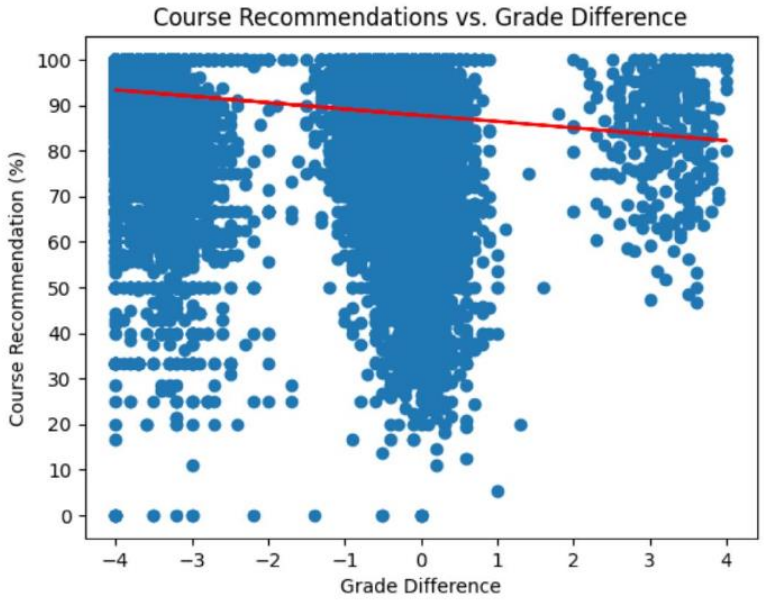
Fig. 2. Survived and not survived number of male and female passengers.

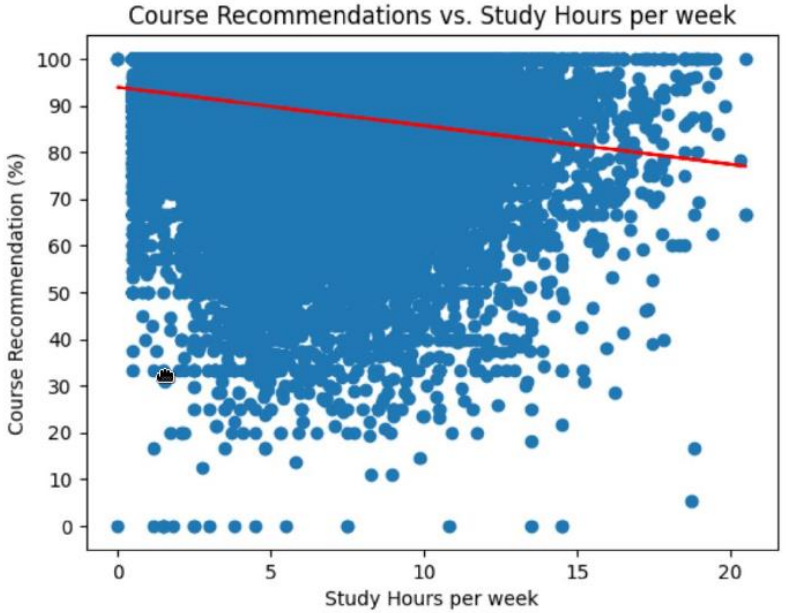


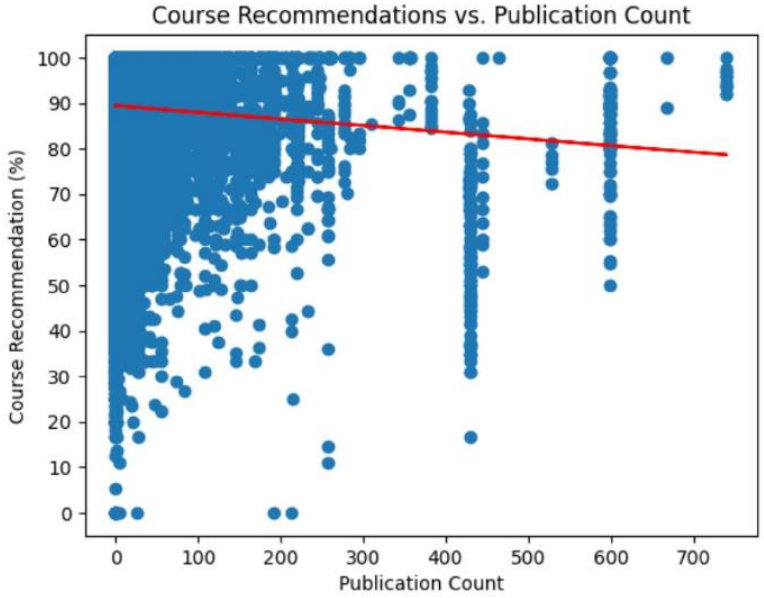


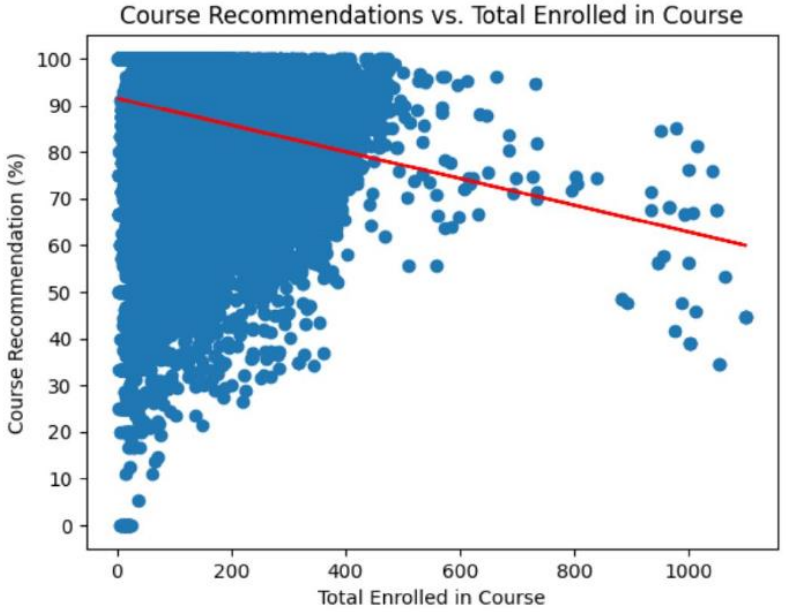
(a) Passenger Class (b) Embark Port

Not all heroes wear capes









Course evals!

Please fill them out!

Other useful feedback?

- Any (useful) suggestions to take-home midterm?
- Is the in-person section "worth it"?
- Textbook useful? Rely on it more or less?
- Peer grading? Keep it forever?
- Other changes?
- Music suggestions?

Thanks!